

PravahAI

Helping Bangluru's traffic congestion via intelligent forecasting and analysis

The Problem

Traffic control rooms find about congestions the same way most of us do: after it has already occurred. This hackathon brief is built by framing the city's traffic disruptions into 2 parts; Planned events like rallies, festivals... and Unplanned events like tree fall, water clogs, accidents...

Our Proposal

Via thorough analysis, we observe that a lot of what gets logged as "unplanned" is not really unplanned but a recurring event which is being missed, and hence creating a gap where the officers have to take action on a sudden basis for an event which could have been foreseen. For example, in the ASTRAM dataset, Mysore Road recorded a heavy vehicle breakdown on Wednesday nights (between midnight and 4:00 AM) in **20 out of the 23 tracked weeks (an 86.9% recurrence rate)**.

This is the gap PravahAI tries to close by taking historical incident log, using an LLM model to find patterns that repeat often enough to be worth planning around and forecasting it with a confidence score. We also implement a "what if" Simulator, a risk calendar and a traffic hotspot cluster with heatmap feature using machine learning on the Bangalore map. Every part will be explained in detail below.

Feature engineering:

- Cleaned the dataset of missing values, lowercasing texts, encoding ID's.
- Derived time features like "day of week". "hour", "month" from the TimeStamp column and converted the UTC values to IST.
- We created 6 time buckets from "hour" column to group events into meaningful traffic period. Namely:

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Bucket	Hours
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Early Morning	5–8
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- Morning 9–12
 - Afternoon 13–16
 - Evening 17–20
 - Night 21–23
 - Midnight 0–4
- We also constructed a corridor-to-zone mapping using a lookup table of Bengaluru administrative zones. This allowed incidents recorded on specific corridors (e.g., Mysore Road, Outer Ring Road) to be aggregated at the zone level for broader spatial analysis. Missing zone values were imputed wherever a corridor-to-zone mapping was available, while corridors with no reliable mapping were retained as "Unknown" rather than being assigned a guessed zone.

After the entire analysis we have 8,139 events and 44 columns divided into 5 parts:

- Core Incident Data
- Geographic & Location Data
- Temporal Features
- Numeric Encodings
- Analytical Risk Scoring

Shadow Events and the SERI score

The core analytical step groups all 8,139 incidents by 3 things at once.

1. Which corridor
2. which day of the week
3. which of the six time buckets

Every combination becomes a candidate "shadow event". When incidents repeatedly occur under the same conditions across multiple weeks, the pattern is flagged as a "shadow event" and assigned a recurrence score indicating how frequently it reappears.

For each of these combinations, we made 3 separate component scores, each normalized to a 0-1 range:

- **Recurrence score:** *"how often does this happen".*
 - *The number of distinct weeks, out of 23, in which that slot saw atleast 1 incident, divided by 23.*
 - $= (\text{Number of distinct weeks in which the slot recorded at least one incident}) / 23$

- **Frequency score:** *"How many times did this incident pattern occur compared to the busiest pattern in the entire dataset?"*
 - $= (\text{Number of events in the slot}) / (\text{Number of events in the busiest slot in the dataset})$

- **Severity Score:** *"How serious is this event pattern compared to the most disruptive pattern in the dataset"*
 - $= (\text{Sum of severity weights of all incidents in the slot}) / (\text{Highest severity-weight sum observed among all slots})$

The severity weight itself comes from a lookup table that assigns a multiplier by cause type: accidents are weighted 3.0, congestion 2.0, water-logging and construction 1.5, tree falls 1.2, potholes and vehicle breakdowns 1.0, VIP movement 0.5, and so on down the list. These are domain judgment calls about how disruptive each cause tends to be.

These categories are logged in the dataset. We logged congestion for a higher weight than others because research shows that a purely volume based congestion requires more manpower and time to untangle rather than a congestion caused by a tree fall/water logging in some part of the road.

- **SERI score:** $= ((0.4 * \text{recurrence}) + (0.4 * \text{frequency}) + (0.2 * \text{severity})) * 100$
 - We invented the Shadow Event Risk Index (SERI) score to pinpoint which events occur where at which time slot and with what severity. This lets us predict optimal manpower and barricading required at the particular event.

 - Recurrence and frequency are weighted equally and heaviest, because a pattern that's both frequent and genuinely spread across many weeks is the strongest signal of something structural rather than a one-off cluster; severity gets a smaller weight because it reflects how bad each incident tends to be rather than how predictable the slot itself is. SERI is then bucketed into four bands:
 - Critical at 81 and above

- High from 61
- Medium from 31
- Low below that

After running this, the dataset produces 666 shadow events.

Corridor and Non-Corridor distinction

We found that about 38% (3,090 of 8,139) events are logged on roads that don't appear on Bengaluru's list of officially named traffic corridors.

Lumping all of those into one catch-all label means "Non-corridor" mathematically dominates every volume-based ranking, since it's effectively one giant bucket competing against dozens of individually-named, narrower ones.

We measured exactly how much this distorts the rankings, and rather than dropping those incidents (which would have thrown away real data) or hiding the skew, we made the dashboard exclude "Non-corridor" from headline rankings by default, with an explicit toggle to bring it back in if you want the unfiltered picture. With non-corridor excluded, the highest-scoring named-corridor pattern is Mysore Road, Wednesday midnight, with a SERI of 43.94.

We also noted that the high number of midnight incidents is supported by the underlying traffic patterns in Bengaluru. Heavy commercial vehicles are restricted from entering the city during the day, causing freight and trucking traffic to be concentrated during late-night hours. Similarly, major civil works such as Metro construction are often scheduled overnight to minimize disruption to daytime traffic.

Among the 1,917 midnight incidents recorded on named corridors, 64% were vehicle breakdowns and 9% were construction-related incidents. Together, these two categories account for nearly three-quarters of all midnight incidents. The corridors with the highest number of midnight incidents, including Mysore Road, Bellary Road, and Tumkur Road, are major routes used by freight vehicles entering and leaving the city.

This suggests that the recurring midnight patterns identified by the system are linked to existing traffic regulations and infrastructure activities rather than random fluctuations in the data.

Hotspot detection (Impact map)

We run the DBSCAN algorithm on the raw latitude/longitude pairs across all 8,139 incidents, with a neighborhood radius of 0.008 degrees and a minimum of 10 incidents required to form a cluster. This produces 14 geographic clusters which are pointed on the map by using MapMyIndia API.

The Forecast Engine

To validate the forecast score(explained below) without letting the model see the future, we hide the final 2 weeks from the model such that the recurrence score and recent-trend-score dont compute it. This is then used as ground truth to check the forecast score against what actually happened. Comparing predicted scores to the ground truth we get a Mean Absolute Score of 0.1375.

- **The forecast score** for a given corridor-day-time bucket slot is $((0.7 * recurrence_Score) + (0.3 * recent_trend_Score))$.
- **Recent_trend_score**(corridor, day_name, time_bucket_ist) = $recent_count(slot) / \max(recent_count \text{ across all slots})$

Similarity Explorer

The Similarity Explorer allows users to select any event and retrieve the 10 most similar events from the 8,139 records in the dataset. Each event is converted into a 42-dimensional feature vector using location, time, event cause, and corridor information, and a Ball Tree nearest-neighbor index is used to find the closest matches.

- Time is represented using a sine-cosine encoding rather than a raw hour value. This avoids treating 11 PM and midnight as being far apart numerically and improves matching for events occurring around the same time of day.
- $Similarity = 1/(1+distance)$
- The resulting score is a measure of relative closeness rather than a percentage. In one validation example, a Wednesday morning pothole incident had a best-match similarity score of 0.59.

The search considers both spatial and categorical information. As a result, events occurring at nearly the same location and time may be matched even if their causes differ, while events with the same cause but occurring far apart may not be considered close matches.

One limitation is the handling of missing corridor information. Events with missing corridor values are assigned the label "unknown," which can cause unrelated events with incomplete data to appear more similar than they actually are.

What-If Simulator

The What-If Simulator allows users to assess a hypothetical event using historical event data. The user selects a location, event type, expected attendance, and day of the week.

- The system then performs two searches. It retrieves historical events within a 2 km radius of the selected location and identifies events with related causes that occurred on the same day of the week.

Using these results, the most frequently affected corridors are identified. Their historical SERI scores are retrieved and adjusted using attendance multipliers of 1.0, 1.3, and 1.7 for small, medium, and large events respectively, producing an estimated risk score capped at 100.

The simulator is intended as a historical reference tool rather than a traffic simulation model. It does not account for factors such as vehicle volumes, signal timings, or route diversions, and the attendance multipliers are heuristic assumptions rather than values calibrated from historical event data.

- For example, selecting Kanteerava Stadium, a Public Event, a Large attendance category, and Saturday causes the system to retrieve 896 historical events within a 2 km radius. Mysore Road appears as the most affected corridor with 304 nearby events. Using the large-event multiplier ($\times 1.7$), its baseline SERI score of 43.9 is adjusted to an estimated risk score of 74.7, indicating a high-risk corridor for event planning.

AI Advisory Layer

Model

- Powered by Gemini 2.0 Flash

Purpose

Converts system outputs and metrics into readable operational advisories for traffic officers.

Supported Advisory Types

- Corridor-level deployment advisories
- Pre-event advisories for planned events

Safety & Reliability Constraints

- Recommendations must be based on actual SERI, recurrence, and forecast data
- Cannot generate unsupported traffic density estimates
- Cannot recommend specific officer deployment counts
- Every advisory is presented as a recommendation requiring human review

Reliability Mechanism

To ensure continuous availability:

- Pre-cached advisories are used for common demo scenarios

- Live Gemini API calls are used for new requests

Learning Engine

Purpose

Tracks forecast performance using operator feedback.

Feedback Collection

For each forecast, operators can submit:

- Accurate
- Partial
- Inaccurate

along with notes describing the actual outcome.

Data Storage

- Feedback is linked to the original forecast record
- All forecast and feedback records are stored for later analysis

Performance Monitoring

- Running error rates can be calculated over time
- Forecast quality can be monitored across different scenarios

Current Limitation

- It currently serves as the data collection infrastructure for a future calibration and improvement loop rather than a self-learning system

The Risk Calendar

It's a 7 by 6 (day by time buckets) grid where each cell takes the highest-SERI event whose day and time bucket match that cell and shows its risk band, the corridor involved and the top recorded cause. A day level summary averages the SERI scores across all of that day's shadow events to give an overall daily risk label.

Effectively, It is a heatmap of the historical pattern density which tells the operator which day and time slot carries elevated risk based on all the data in our database(aka knowledge base).

Backend

- Built using **FastAPI**
- Exposes **23 API routes**

Offline Processing Pipeline

The following computations are performed once during preprocessing:

- Data cleaning
- Feature engineering
- Shadow event scoring
- DBSCAN hotspot clustering
- Forecast generation

These artifacts are loaded into memory when the API starts, allowing most requests to return precomputed results without performing expensive calculations in real time.

Database

Storage Engine

- SQLite

Tables

- Raw incidents/events
- Shadow events
- Hotspots
- Forecast log
- Learning feedback log

Frontend

Technology Stack

- React
- TypeScript
- Vite

Mapping and Visualization

- Leaflet for interactive maps
- CARTO Dark Tiles as the map basemap
- Leaflet Heat Layer for hotspot visualization
- Recharts for charts and dashboards

Known Limitation

- MapMyIndia tiles were originally planned but required a domain-whitelisted API key that was unavailable.

- CARTO tiles were used as an alternative.

Application Modules

The platform consists of approximately 12 pages:

Analytics & Monitoring

- Dashboard with KPIs and charts
- Operational map
- Risk calendar
- Forecast view

Pattern Discovery

- Shadow events table
- Similarity explorer
- Hotspot visualization

Decision Support

- What-If simulator
- Learning engine feedback page

Knowledge Management

- Knowledge repository
 - Full-text search
 - Seven filtering options
 - Access to all 8,139 historical events